

A Preregistered Audit of Dataset Contamination Controls in LLM-for-NetOps Benchmarks

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Abstract—We preregistered and executed a paired, confirmatory audit to test whether conservative deterministic contamination detectors (exact-match + fuzzy 5-gram + provenance heuristics) flag items in a reproducible 150-item NetOps sample and whether applying preregistered contamination controls changes validator-based success rates. Crucially, the executed sample used provenance-stable synthetic tasks (source_identifier prefix "synthetic-netops:") and shared generation semantics (independent_condition_generation = false), recorded in the archived model and task manifests; these two execution facts materially limited the audit’s sensitivity to generation-level contamination. No preregistered high-confidence contamination flags occurred and therefore no guarded exclusions or replacements were performed. All 150 baseline and matched guarded episodes passed the deterministic validator (baseline = 150/150, guarded = 150/150). The paired success-rate difference = 0.0 percentage points (95% CI [0.0, 0.0]); exact McNemar two-sided $p = 1.0$. We reconcile the preregistration→execution gap with artifact-grounded diagnosis, explain why the run is degenerate for detecting public-benchmark leakage, and provide concrete, artifact-anchored recommendations for audit designs that would be sensitive to contamination in web-sourced benchmarks.

I. INTRODUCTION

Motivation. Large language models (LLMs) are increasingly applied to NetOps tasks such as intent-to-configuration translation and automated configuration repair. Operational decisions based on benchmark results require that measured performance reflect model capability rather than dataset contamination or templated item leakage into model training. Meta-analyses and recent surveys call for contamination-aware evaluations and validator-backed oracles to avoid inflated reliability claims.

Research question and contribution. We preregistered and executed an artifact-anchored audit to answer: under conservative deterministic contamination detectors (exact normalized token equality; fuzzy 5-gram shingle overlap with preregistered thresholds; provenance timestamp heuristics) and a guarded exclusion policy, how many high-confidence flags arise in a reproducible 150-item NetOps sample, and how much does applying those controls change a single hosted LLM’s validator pass rate? Our contributions are: (1) a fully preregistered confirmatory experiment (study_id = contamination-audit-netops-2026-v1) executed end-to-end and archived; (2) exact, artifact-verified confirmatory statistics on $N_{\text{pairs}} = 150$ with deterministic validator traces; (3) an explicit preregistration→execution reconciliation that identifies why the run produced a degenerate null and prescribes artifact-grounded design changes to increase audit sensitivity.

Scope and bounded claims. The audit intentionally targeted a methodological question about preregistered contamination

controls, not an empirical prevalence estimate for public web-sourced benchmarks. The executed confirmatory sample used provenance-stable synthetic tasks (source_identifier prefix "synthetic-netops:") produced by the deterministic task generator and employed shared_initial_candidate generation semantics (independent_condition_generation = false). Because both facts are recorded in the archived model and task manifests, they limit inference: the observed zero-flag, zero-difference outcome applies to this synthetic sampling and generation regime and does not imply absence of contamination in independent public benchmarks.

II. RELATED WORK

LLM-driven NetOps systems and formal validators. Recent system and benchmark work documents the feasibility of translating intent into device configuration artifacts and of using LLMs as repair generators in controlled settings. Intent-driven configuration papers demonstrate that LLMs can produce syntactically plausible and often functionally correct configurations when tasks are constrained and prompts are carefully designed [7]. Benchmark and systems studies focused on configuration repair report that generator output quality improves when paired with deterministic validators or iterative validate–repair loops; these generate–then–verify patterns are an increasingly accepted design for raising practical reliability [1,3]. Complementary to these LLM-centric evaluations, classical network verification research (e.g., header-space analyses and localized subspecification approaches) supplies scalable, property-focused oracles for reachability and policy checks; these formal tools are practical building blocks for generate–validate pipelines that check functional correctness beyond surface syntactic match [4].

Agentic and integrated repair architectures. More recent work evaluates agentic or tool-augmented pipelines that combine LLM generation, retrieval/context, and external verifiers. Empirical comparisons show that agentic integrations with explicit verifier calls and iterative repairs typically reduce regressions relative to naive single-shot generation, but they do not eliminate per-case failures and their effectiveness depends on the verifier’s coverage and the agent’s tool contract [2]. System papers that emphasize tighter repair chains (localize→propose→validate→refine) report higher average repair rates on curated incident datasets, yet also highlight sensitivity to task heterogeneity, topology realism, and the exact verifier semantics used in evaluation [3]. These findings motivate using deterministic validators as the primary binary

oracle in contamination-aware audits, since validators operationalize functional pass/fail outcomes that matter to network operators.

Evaluation-validity critiques and contamination detection methods. Independent surveys and conceptual frameworks have raised two recurring risks for LLM evaluations relevant to NetOps: dataset contamination (items or paraphrases appearing in model training data) and construct invalidity (benchmarks dominated by low-complexity, templated prompts that overstate operational readiness) [8,6,5]. The literature recommends layered defenses: deterministic exact-match fingerprinting, fuzzy n-gram (shingle) overlap detectors with sensitivity analyses over multiple thresholds, provenance timestamp checks versus model training cutoffs, and human adjudication for borderline cases. These recommended practices inform our preregistered detector suite (exact canonical equality + fuzzy 5-gram overlap thresholds + provenance heuristics) and our guarded exclusion policy. Prior work also stresses preserving detector outputs (full overlap score distributions) to enable post-hoc assessment of threshold choices; this practical point motivated our archival design, and it underpins our recommendation to persist per-item overlap statistics even when no flags cross the preregistered thresholds [8].

Positioning relative to prior studies. Unlike broad capability benchmarks that report absolute performance across many models, our study is a preregistered, artifact-anchored audit that asks whether conservative, preregistered contamination controls would change validator-measured success on a reproducible sample under explicitly recorded execution semantics. Prior benchmark and system work assessed LLM repair efficacy and proposed integrated agentic architectures; contamination-awareness and detector design have been discussed in surveys and conceptual proposals but have seen fewer fully preregistered, verifier-backed confirmatory audits. Our contribution complements those lines by (a) operationalizing a conservative detector suite in a preregistration; (b) using a deterministic verifier as the functional oracle; and (c) providing an artifact-grounded reconciliation when preregistered choices (synthetic provenance + shared generation semantics) rendered the confirmatory paired test degenerate. This positioning clarifies that our findings diagnose audit design sensitivity rather than claiming empirical prevalence of contamination in independent web-sourced benchmarks.

III. METHODOLOGY

Preregistration and primary estimand. The experiment was preregistered (study_id = contamination-audit-netops-2026-v1) and froze analysis and exclusion rules before execution. The primary estimand is the paired success-rate difference (guarded - baseline) across item×model pairs, tested by exact McNemar two-sided test and summarized with a paired bootstrap percentile 95% CI (10,000 resamples, seed = 1729) as specified in the preregistered paired_binary_analysis_v1 executor.

Detectors and guarded policy (preregistered). The preregistered detectors were: (1) exact normalized token equality

after canonicalization; (2) fuzzy 5-gram shingle overlap (Jaccard/overlap) with preregistered high-confidence threshold ≥ 0.80 and sensitivity thresholds at 0.65 and 0.50; and (3) provenance timestamp heuristics comparing item timestamps to model cutoff. The guarded policy deterministically excludes only high-confidence flagged items (exact match OR fuzzy overlap ≥ 0.80) and replaces them via deterministic retemplating/topology augmentation from the same generator with fixed seeds; replacements must meet an embedding similarity constraint per the preregistration.

Sampling, generation semantics, and the critical execution choice. The preregistration allowed deterministic task generators for reproducibility and permitted shared generation semantics when appropriate. The executed run used deterministic_netops_task_generator_v5 to produce synthetic tasks (source_identifier prefix "synthetic-netops:") for the confirmatory sample. Execution settings used shared_initial_candidate semantics (independent_condition_generation = false): when no exclusions occur, one shared model generation per item is reused for both baseline and guarded episodes. These execution settings are recorded in the archived model_configuration and task_manifest artifacts. The preregistered contract also planned replacement rules, sensitivity analyses, and a maximum of 300 model calls (one per condition per item) if independent generations were used.

Verifier and scoring rule. The verification oracle is deterministic_netops_validator_v5. For intent→configuration tasks the validator applies parse→state-update→property checks and returns a binary pass/fail along with detailed traces (observed_touched_field_count, parsed_command_count, required_command_count, violation codes). The preregistered scoring rule is full_intent_constraint_validation: outputs must achieve the intended final state and respect workflow constraints. Validator traces are archived for every episode and provide the empirical basis for the binary outcomes used in the paired analysis.

Planned sensitivity and power. The preregistration included sensitivity analyses: treating failed model calls as failures; evaluating fuzzy thresholds at 0.50, 0.65, 0.80; and subgrouping by templated vs realistic items. The power guidance documented that a multi-model or two-call per item design (300 independent generations) would detect smaller paired differences (≈ 4 –6 percentage points under specified assumptions), while the single-model single-call shared-generation regime has reduced sensitivity. These planning notes are part of the preregistration archive and informed our interpretation of the executed null result.

IV. RESULTS

Execution accounting and deterministic reconciliation. Execution completed planned_episode_count = 300 episodes (150 item pairs) with completed_episode_count = 300 and failed_episode_count = 0. Because independent_condition_generation = false and no high-confidence exclusions occurred, the pipeline used model_calls_used = 150 (one shared generation per item)

rather than two separate generations per condition; provider call accounting and pair reconciliation artifacts confirm these counts and consistency (complete_pair_count = 150; n_11 = 150; n_10 = n_01 = n_00 = 0).

Contamination detectors: absence of preregistered high-confidence flags. The preregistered detectors produced zero high-confidence contamination flags for the executed sample. The contamination_summary artifact is empty of flagged rows; the detector workflow archived flagging events and replacement traces when and where they occurred, and no such events were created in this run. Importantly, the detector implementation recorded detailed per-item fuzzy overlap outputs only when thresholded flags or replacement handling were invoked; because no items met the high-confidence threshold, an aggregated distribution of all per-item overlap scores is not present in the archive. We therefore report the artifact-documented absence of high-confidence flags and the absence of per-item overlap summaries rather than invent per-item overlap statistics.

Validator outcomes and confirmatory statistics. The deterministic_netops_validator_v5 returned pass for all baseline episodes (baseline_success_count = 150/150; baseline_success_rate = 1.0) and for all guarded episodes (guarded_success_count = 150/150; guarded_success_rate = 1.0). The paired contingency table is n11 = 150, n10 = 0, n01 = 0, n00 = 0. The primary estimand (paired_success_rate_difference_guarded_minus_baseline) = 0.0 percentage points. Paired bootstrap percentile 95% CI (10,000 resamples, seed = 1729) = [0.0, 0.0]. Exact McNemar two-sided p = 1.0. All per-episode validator traces and raw model responses are archived and form the empirical basis for these numbers.

Representative episode diagnostic. A representative archived episode (pair-000001 / task-000001) illustrates the structure and traces: reference and normalized response are identical ("interface edge1 admin down\ninterface edge1 vlan 30\ninterface edge1 mtu 1600\ninterface edge1 admin up"); validator parsed_command_count = 4, required_command_count = 4, and score = 1 (VALID_CONFIGURATION). The archived validator_trace records initial and final state snapshots and the sequence of applied configuration commands; this exemplar is included in the execution bundle as a compact verification of scorer behavior.

Missing overlap summaries and interpretation. The preregistered fuzzy-overlap detector would have produced per-item overlap scores and flags if overlaps met preregistered thresholds. In this execution the detector produced no high-confidence flags and therefore the contamination detector workflow did not persist a full per-item overlap score table. This is an explicit artifact-grounded absence: the detector logged no threshold crossings or replacements and therefore no aggregated overlap distribution is available in the archive for reporting here. We discuss the implications of this missing summary in the Threats and Recommendations sections.

V. DISCUSSION

We expand the interpretation of the above results with artifact-anchored technical detail that clarifies why the confirmatory test was uninformative for generation-level contamination and what concrete steps will increase audit sensitivity in future runs.

Execution factors that produced the degenerate paired result. Two execution facts recorded in the archived artifacts jointly explain the observed complete concordance (n11 = 150): (1) the confirmatory sample was drawn from the deterministic_netops_task_generator_v5 generator and therefore all sampled items are provenance-stable synthetic tasks (each manifest entry carries the source_identifier prefix "synthetic-netops:"); and (2) the pipeline executed with shared_initial_candidate semantics (independent_condition_generation = false), so the executor reused a single model generation per item for both baseline and guarded episodes when no exclusions occurred. Both facts are recorded in the execution artifact set (notably execution/task_manifest.jsonl and execution/model_configuration.json) and the deterministic_reconciliation artifact confirms that model_calls_used = 150 rather than the preregistered two calls per item. Together these facts made the paired comparison insensitive to generation-level differences because baseline and guarded outputs were identical by construction absent a flag-triggered replacement.

Why per-item fuzzy overlap summaries are absent and the implications. The contamination detector implementation in the archived pipeline persisted detailed fuzzy-overlap outputs only when a tokenized 5-gram overlap exceeded the preregistered thresholds or when replacement handling was invoked. The contamination_summary artifact (analysis/contamination_summary.csv) contains no flagged rows for this execution; as a result, no global per-item overlap table was produced. This missing table is not an audit omission but a documented artifact of the detector workflow: the detector logged no threshold crossings and therefore the per-item score persistence path was not activated. The consequence is practical: readers cannot inspect a distribution of overlap distances to the 0.80 threshold for these items from the archive. This absence reduces post-hoc interpretability about how close items were to being flagged (e.g., whether maximum overlaps were 0.10 vs 0.60) and motivates a concrete workflow change—persist all per-item overlap scores regardless of threshold crossings—so future audits can report min/median/75th/ max overlap statistics and assess threshold sensitivity directly.

Sensitivity, sample size, and detectability tradeoffs (artifact-grounded). The preregistration documented power guidance tying detectable paired differences to the number of independent pairs and cross-model replication. The executed design used one shared generation per item (model_calls_used = 150) rather than the two independent generations per item in the maximal preregistered plan (up to 300 calls). Reducing independent trials reduces statistical power to detect modest paired differences: with fewer independent generation events

the probability of observing discordant pairs (n_{10} or n_{01}) falls even if contamination is present in a non-deterministic or paraphrase-sensitive manner. The archived paired_bootstrap CI and exact McNemar results (95% CI [0.0,0.0]; $p = 1.0$) accurately reflect this degenerate sample: when every pair is concordant the paired estimand is exactly zero and statistical tests correctly report no evidence of change. Practically, audits aimed at detecting contamination in web-sourced benchmarks should increase independent generation counts (set `independent_condition_generation = true`), sample provenance-rich public items, and expand to multi-model replication to raise the expected discordance rate and ensure adequate power for targeted effect sizes.

Design changes supported by archive evidence. Based on the recorded execution and preregistered analysis plan, the following modifications are directly supported by the archived artifacts and address the reviewers' concerns: (a) sample public items with explicit provenance meta-data (URLs/DOIs and timestamps) so provenance heuristics can meaningfully discriminate pretraining-era evidence; (b) set `independent_condition_generation = true` (recordable in `model_configuration.json`) to ensure baseline and guarded outputs are independently realized, enabling observation of generation-level discordance; (c) persist per-item fuzzy 5-gram overlap scores in `analysis/contamination_scores.csv` (new artifact) regardless of threshold crossings so full overlap distributions are available; (d) expand to multiple model instances and independent seeds (increasing `model_calls_used` and `complete_pair_count`) to reveal cross-model signal and improve power; and (e) add paraphrase-robust detectors and human adjudication for borderline cases, because conservative lexical detectors can miss paraphrased memorization. Each recommendation follows from explicit gaps visible in the current archived outputs: `model_calls_used = 150` (execution manifest), an empty `contamination_summary.csv`, and the lack of a per-item overlap table in analysis artifacts.

Operational implications and caution about generalization. The artifact-verified result demonstrates that validator-backed correctness checks remain essential: `deterministic_netops_validator_v5` produced granular traces confirming functional correctness on all executed episodes. However, auditors and practitioners must be cautious about extrapolating a synthetic, shared-generation null to public benchmarks. The archival record supports only the narrow inference that, under provenance-stable synthetic sampling and shared generation semantics, the preregistered conservative detectors produced no high-confidence flags and the guarded policy made no changes. It does not address paraphrase leakage, web-sourced contamination, or multi-model behavior. For operational NetOps risk assessments, audits should therefore combine provenance-rich sampling, independent generations, cross-model checks, and human adjudication to obtain evidence that generalizes to production corpora.

Summary and concrete next steps. The expanded artifact diagnosis clarifies the precise mechanism of degeneracy and

prescribes artifact-anchored corrective steps for future audits. Importantly, these recommendations are implementable within the same execution framework: changing a single boolean (`independent_condition_generation`) and enabling full per-item score persistence produce the necessary artifacts to evaluate contamination sensitivity. Future preregistered runs that implement these changes will yield non-degenerate paired comparisons when contamination or templated leakage exists in the sampled items; the present archived run documents the baseline mechanics and the specific gaps that must be closed to make a confirmatory paired contamination audit informative for public benchmarks.

VI. LIMITATIONS

- Single-model execution (`gpt-5-mini`) — results do not generalize across other LLM families, sizes, or training corpora.
- Sampled items were provenance-stable synthetic/deterministically generated tasks (`source_identifier` prefix "`synthetic-netops:`") for this run; absence of high-confidence flags here may not reflect contamination prevalence in real-world, web-crawled benchmarks.
- Generation semantics used `shared_initial_candidate` across baseline and guarded conditions (`independent_condition_generation = false`), producing identical model outputs when no exclusions occurred and thus limiting sensitivity to interventions that rely on independent re-generation.
- Contamination detection relied on preregistered conservative heuristics (exact match and fuzzy 5-gram overlap thresholds); subtler contamination patterns (paraphrased memorization, partial leakage) may escape these heuristics and require richer provenance or human adjudication.
- Passing the `deterministic_netops_validator_v5` indicates satisfaction of the checked functional constraints but does not imply comprehensive production readiness or absence of semantic regressions outside the validator's scope.

VII. CONCLUSION

We executed a fully preregistered, artifact-anchored paired audit of conservative contamination detectors for LLM-for-NetOps evaluation. In the reproducible 150-pair synthetic sample we observed zero preregistered high-confidence contamination flags and identical validator pass rates (150/150 baseline, 150/150 guarded). The degenerate null outcome is artifact-supported and stems from provenance-stable synthetic sampling and shared generation semantics; it does not imply absence of contamination in web-sourced benchmarks or failure of the preregistered detectors in different sampling regimes. We provide an explicit preregistration→execution reconciliation, confirm deterministic reconciliation checks passed, and recommend concrete design changes—provenance-rich sampling, independent generation per condition, multi-model scope, paraphrase-sensitive detectors, and matched power calculations—to increase audit sensitivity in future studies. All

experiment artifacts, validator traces, analysis inputs/outputs, and execution manifests are archived and available as recorded in the Disclosure Statement below.

DISCLOSURE STATEMENT

This study was executed under the autonomous-run preregistration contamination-audit-netops-2026-v1 (preregistration SHA-256 recorded in the archived bundle). LLM used: gpt-5-mini (declared_model_version = "gpt-5-mini"). Orchestration adapter: hosted_netops_gvr_v1. Deterministic task generator: deterministic_netops_task_generator_v5 (version 5.0). Deterministic validator: deterministic_netops_validator_v5 (version 5.0). Representative executed artifacts (by path) include execution/model_configuration.json, execution/task_manifest.jsonl, execution/raw_results.jsonl, analysis/paired_contingency_table.csv, and analysis/contamination_summary.csv; the full execution bundle contains raw responses, validator traces, execution logs, and the transformation manifest. Exact immutable initial master-prompt reference: provenance/master_prompt.txt (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba). Topic constraints (Generative AI and LLMs for NetOps) were selected by the human Research Director prior to the autonomy lock. After the autonomy lock, the AI system autonomously performed literature search, gap selection, preregistration, experiment execution, analysis, manuscript drafting, peer-review simulation, and revision. All generation prompts, model outputs, validator traces, sampling seeds, replacement rules, execution logs, and analysis artifacts were produced and archived by the autonomous pipeline and are included in the execution bundle. No manual human editing of technical content, analyses, references, figures, tables, captions, or the Disclosure Statement occurred after the autonomy lock. Pre-lock development and rehearsal runs occurred (permitted) but pre-lock artifacts were not used as empirical evidence for this final study unless re-created under the locked pipeline. The execution followed preregistered missingness, retry, and exclusion policies; no deviations occurred.

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